

INVESTIGATION ANALYTICS PLATFORM

Crime Investigation Analytics

Los Angeles | 2020–2024

Power BI · Advanced DAX · 800K+ Records

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Project Objectives



Map Crime Hotspots

Surface high-concentration crime zones across all 21+ LAPD divisions.



Analyze Trends

Track year-over-year and seasonal shifts across five years of data.



Surface Patterns

Break down offense types, weapons, and victim demographics at a glance.



Drive Decisions

Turn raw incident data into evidence-based actions for resource and policy decisions.

THE PROBLEM

How do you find meaningful connections inside more than 800,000 crime records?

Thousands of incidents across 21+ divisions

Overlapping offenses and case relationships

Patterns hidden inside five years of data

THE APPROACH

Raw data becomes investigation-ready intelligence.

Raw Crime Data

Power Query · cleaning



Data Model

Snowflake schema · DAX



Investigation Analytics

Geographic · temporal · relational



Actionable Insights

Evidence-based decisions

Dataset Overview

800K+ Crime Records

Incident-level data sourced directly from the LA Open Data Portal.

2020–2024 Analysis Period

Five full years of reported crime across Los Angeles.

21+ LAPD Divisions

Complete geographic coverage of all LAPD reporting areas citywide.

30+ Crime Categories

From violent offenses to property and quality-of-life crimes.

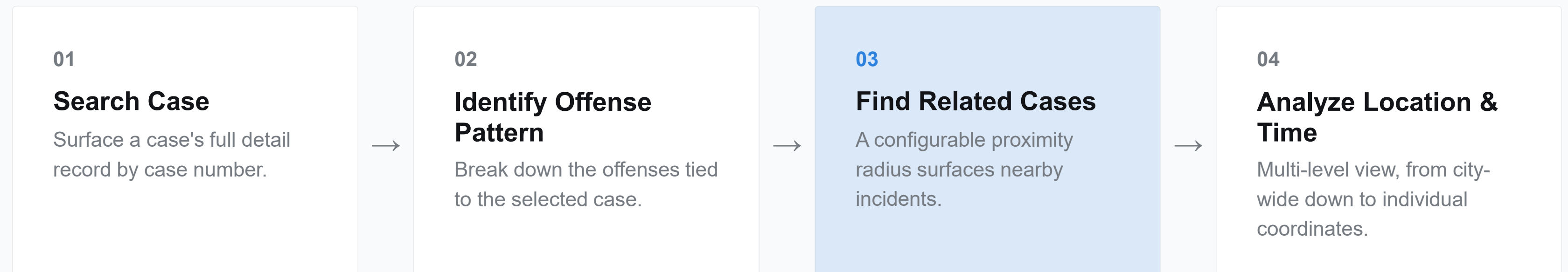
15+ Attributes per Record

Date, location, weapon, victim demographics, and case status — per incident.

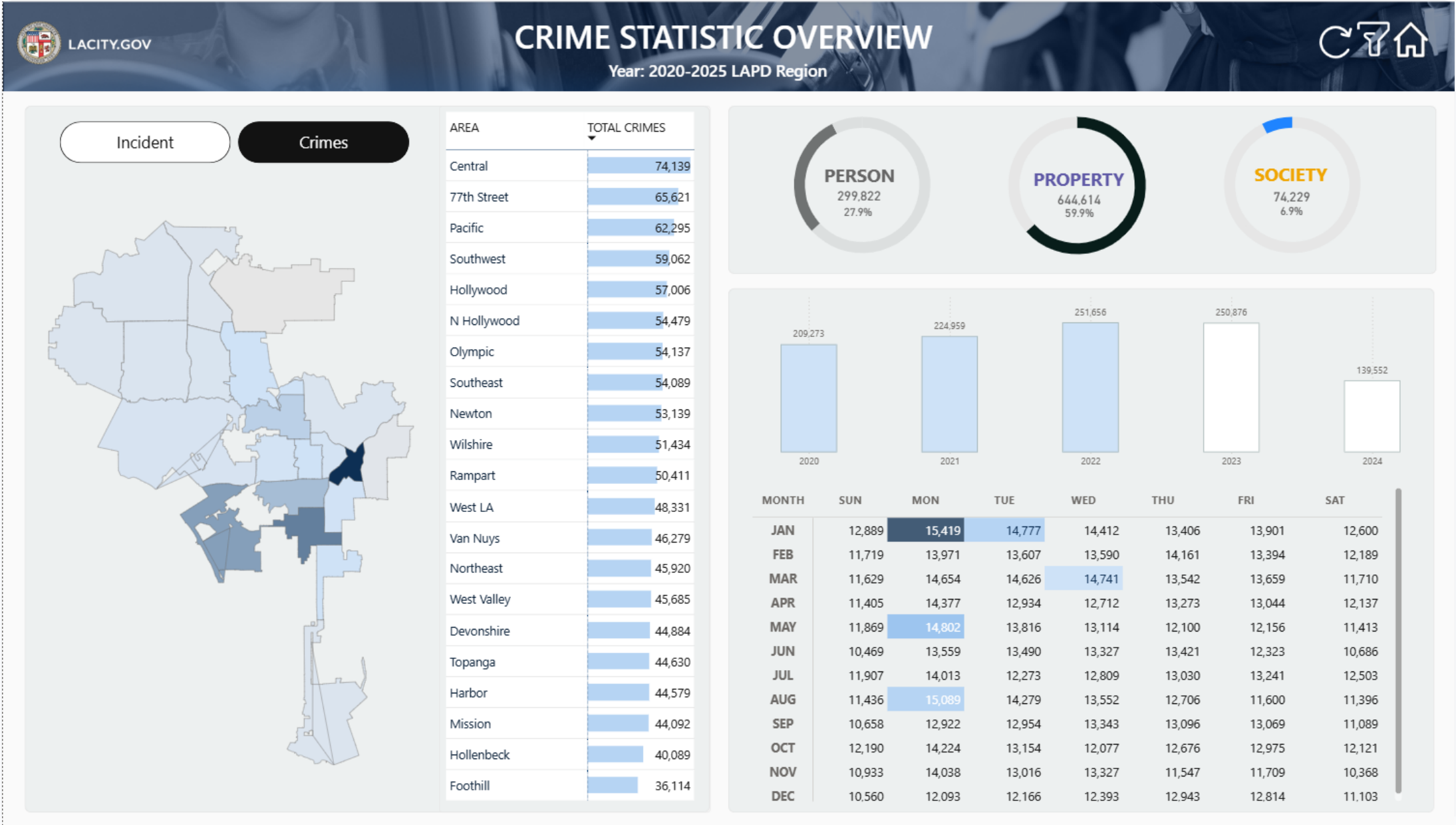
Source: LA Open Data Portal

Publicly available, updated regularly — real-world data, not synthetic.

Investigators move from a single case to related cases in seconds.



Executive Dashboard: Page 1 Overview



Interactive LAPD Map

Surfaces geographic crime hotspots across divisions in real time.

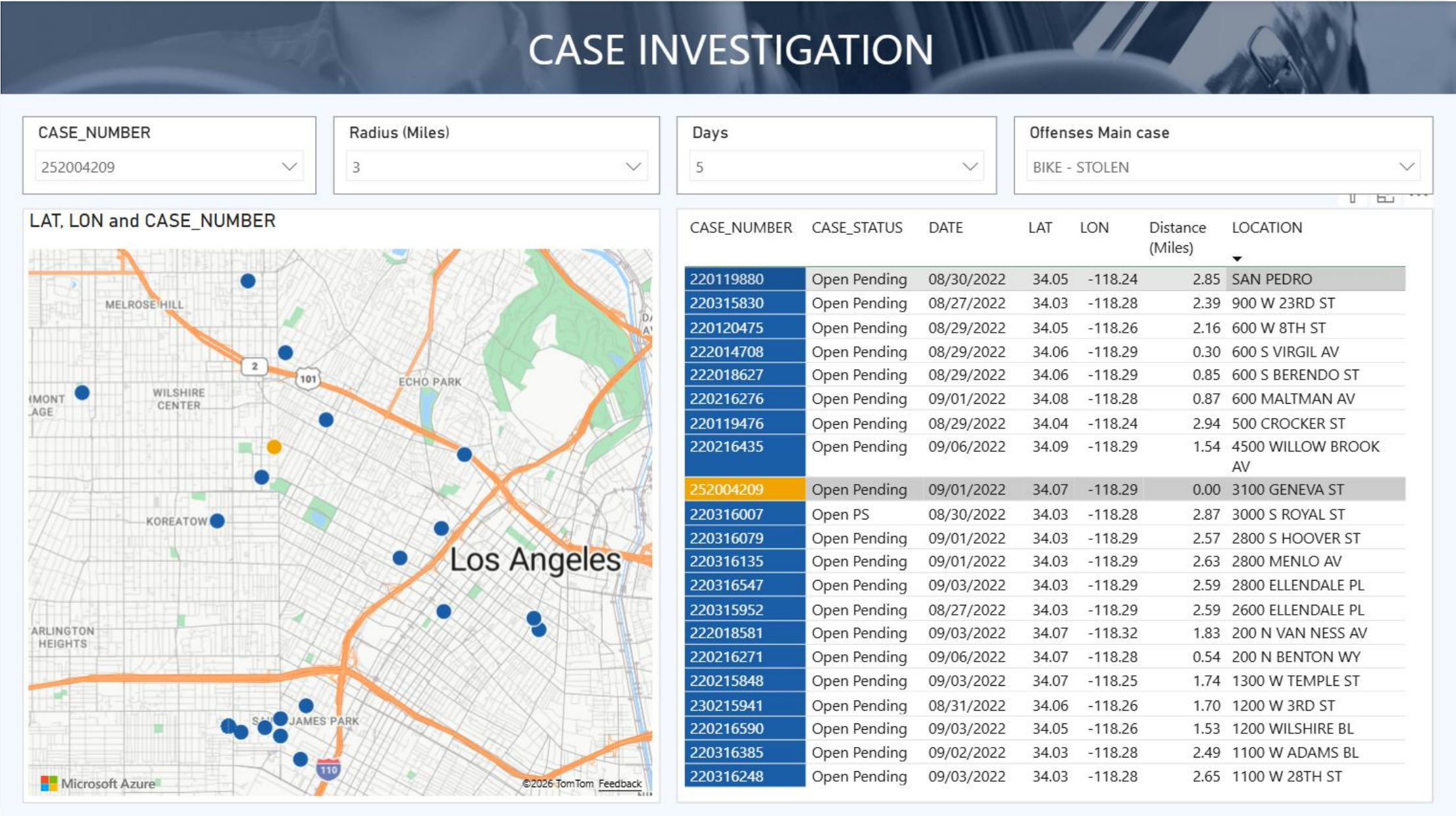
KPI Cards

Crime volume broken down by persons, property, and society categories.

Trend & Heatmap

Year-over-year chart and monthly heatmap reveal seasonal patterns from 2020–2024.

Interactive Case Investigation



Key Insights Uncovered

Crime peaked in 2022

Incident volume hit its highest point post-pandemic, then declined — a trend invisible without five-year data.

77th Division leads by volume

The highest-crime LAPD division, consistently — a signal for resource prioritization, not just observation.

Friday evenings are highest-risk

Weekly heatmaps show a clear spike every Friday 6–10pm across most crime categories.

Vehicle theft is the #1 offense

Growing year-over-year while violent crime declined — two different stories in one dashboard.

Value



2020 2021 2022 2023 2024

Category

Business Recommendations

Four evidence-based actions derived directly from the data.

1

Redeploy to 77th Division

Highest incident volume in the city — prioritize patrol resources and accelerate response times.

2

Target Friday Evenings

The peak crime window is consistent and predictable — schedule surge capacity before it hits.

3

Tackle Vehicle Theft

Fastest-growing offense category. Targeted deterrence programs can reverse the trend.

4

Monitor Post-2022 Decline

Crime fell post-2022. Track leading indicators now to sustain and accelerate the decline.

Skills Demonstrated



SQL

Data extraction, filtering, and aggregation from relational databases.



Power Query

End-to-end data cleaning, transformation, and loading into the BI model.



Data Modeling

Snowflake schema design with optimized fact and dimension table relationships.



DAX

KPIs, year-over-year measures, and dynamic context-aware filtering built from scratch.



Power BI

Multi-page interactive dashboard with drill-through, cross-filtering, and live KPI cards.



Data Storytelling

Translating 800K+ records into insights executives can understand and act on.

From raw crime records to decisions that matter.

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